

Introduction to Using the Arduino Development System

ref1: <https://docs.arduino.cc/hardware/uno-rev3/>

Instructor Notes

0. Setting up the Arduino IDE

Direct the Students to the following link for the Arduino website.

(see <https://www.arduino.cc/> > For Makers > Go to Documentation > Software > Arduino IDE 2)

It has all of the information needed and will be a valuable resource.

1. Blink; the hardware “Hello World” application.

Have the students load the Blink program from “File > Examples > 01.Basics > Blink” and upload in to their Arduino board. The most common problem is having the wrong COM port selected. Use the Board Select pull-down in the Tab bar to verify the correct COM port is selected. Make sure that they Save the project into their own custom folder (usually in Documents) BEFORE editing the Blink file (it’s usually READ-ONLY). They may have to change the file Properties (unclick the READ-ONLY attribute).

2. LEDs – Some electronics theory is now needed so that parts (the Arduino or the LEDs) are not damaged. This section will cover Ohm’s Law, a little bit about diode properties, and the Arduino interface hardware. The Atmel328P (the processor used in the UNO) data sheet shows that the device can sink and source 20ma. Kirchoff’s Voltage Law (KVL), Kirchoff’s Current Law (KCL), and Ohm’s Law are used to calculate the series Resistor value used with a standard red LED.

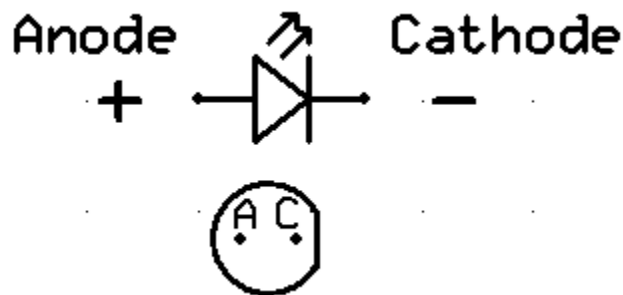


Figure 1 - LED

Other common resistor values are mentioned if the calculated value is not available.

Suggestion: Have an assortment of resistors available for the students to see the affect current on LED brightness.

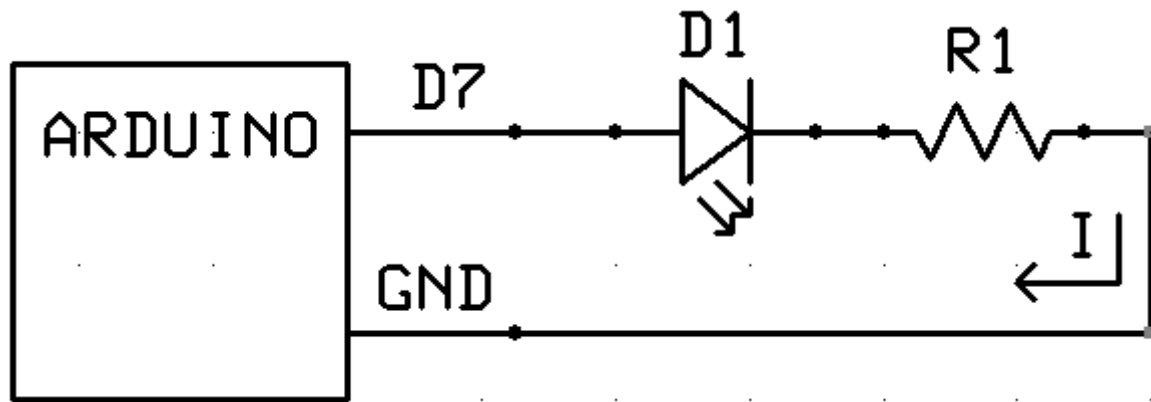


Figure 2 – Simple Arduino circuit

3. Adding external LEDs to the Arduino aboard. Apply information from #2.

Experiments with six LEDs:

3a. Turn each LED ON; one at a time, then turn each LED OFF, one at a time.

3a1. Straight line code.

3a2. Using as **Array** with an ON function and then an OFF function.

3a3. Using an Array of **Data Structures** {LED#, State} and one function.

Use the Array as a circular pattern generator.

Use the Array as a scrolling patten generator; Chase, Ping-pong, Morse Code.

3a4. Using the random function to randomly pick an LED from the Array to turn ON/OFF or Toggle.

Use an Array to record the **State** of an LED.

3b.